

NPTEL Course on "Computer System Optimization for ML"

Lecture #30

Agenda

- Recap
- Orca (LLM Serving)

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Summary

- **Key idea:** Make KV cache from a collection of small tensors (e.g., 16 tokens)

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Slides on vattention borrowed from Ashish Panwar

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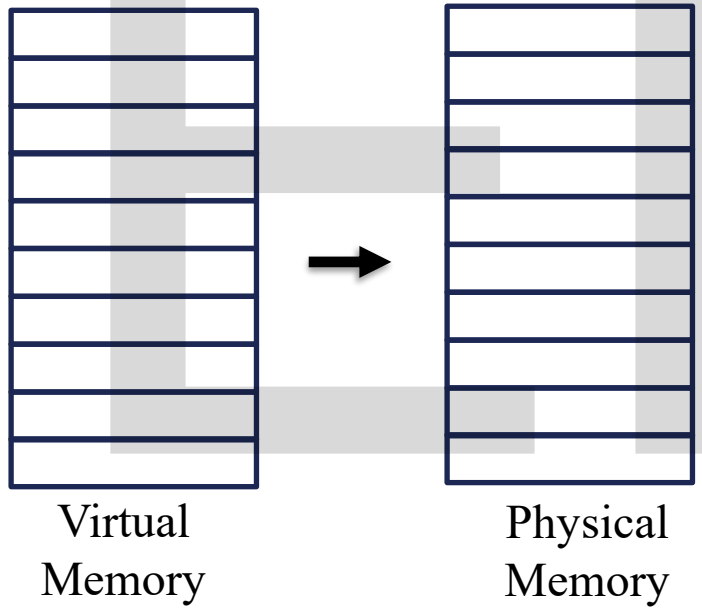
KV cache is dynamic.

Slides on vattention borrowed from Ashish Panwar

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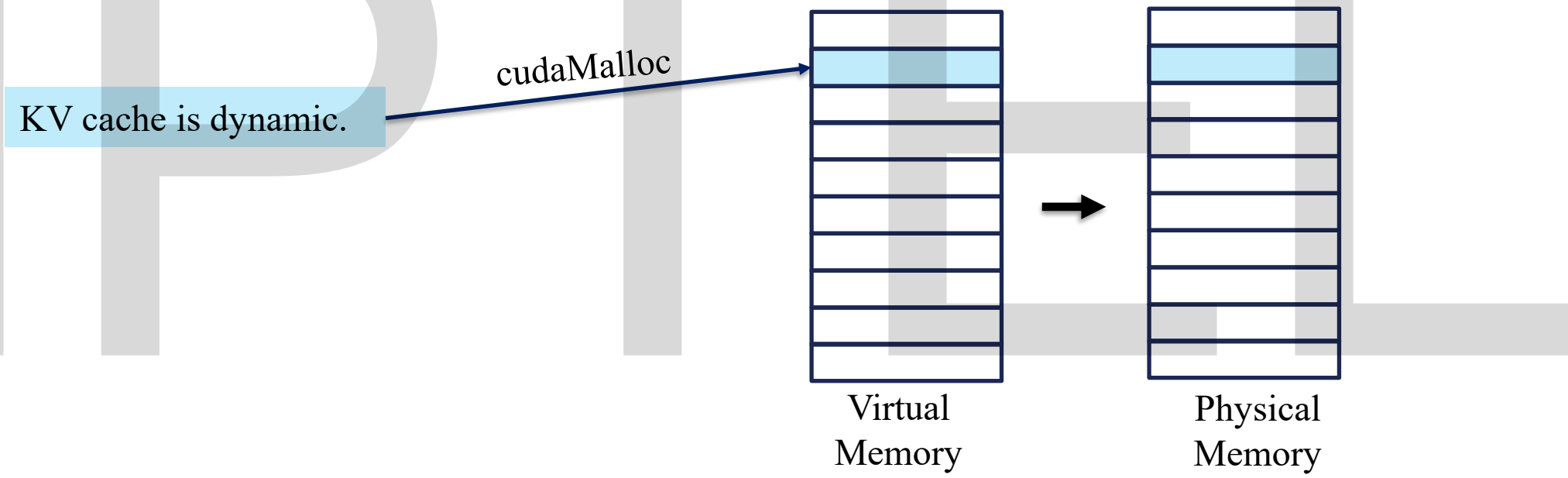


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Slides on vattention borrowed from Ashish Panwar

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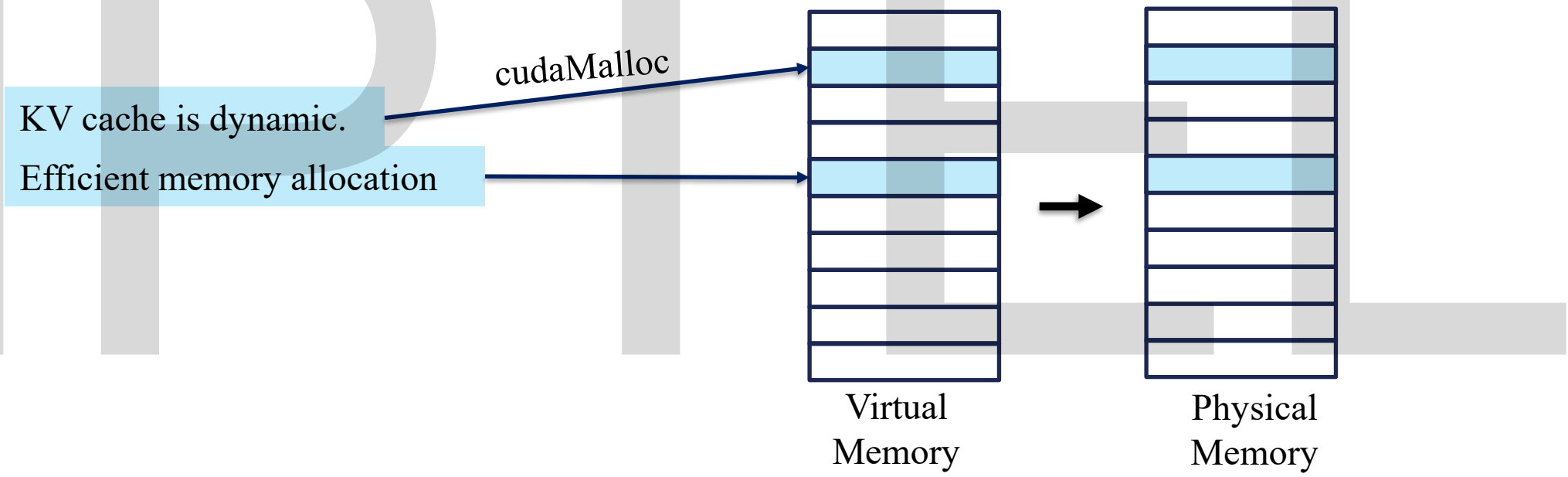
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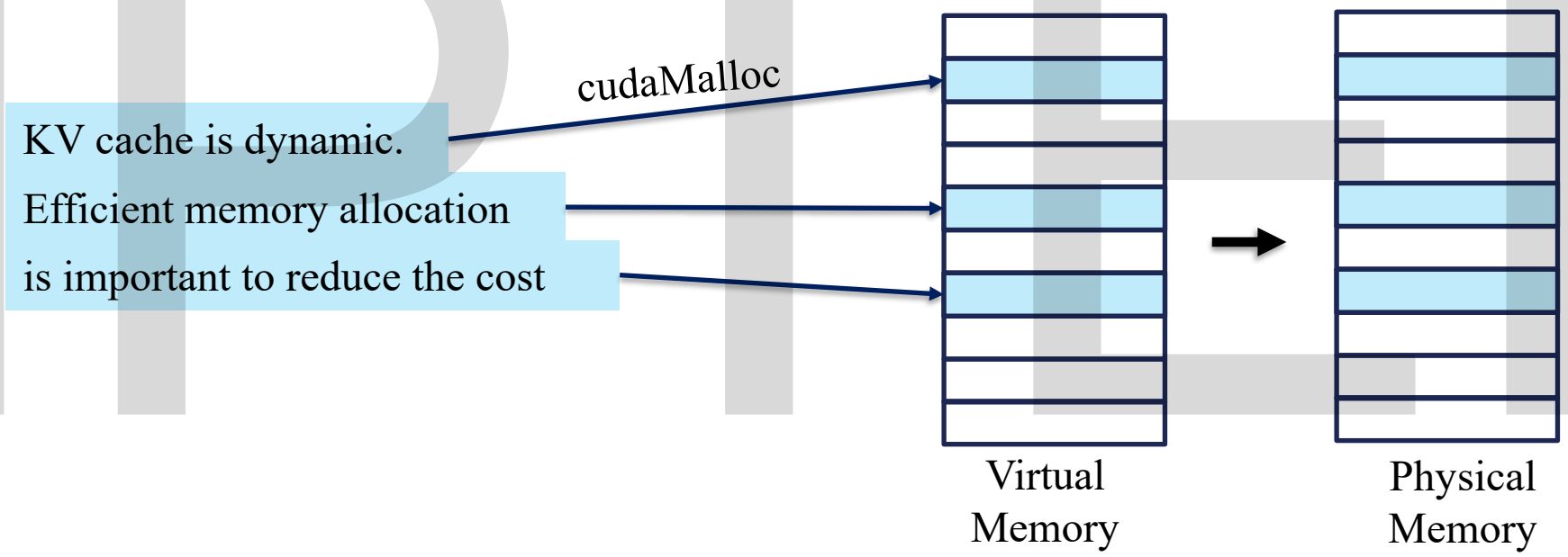
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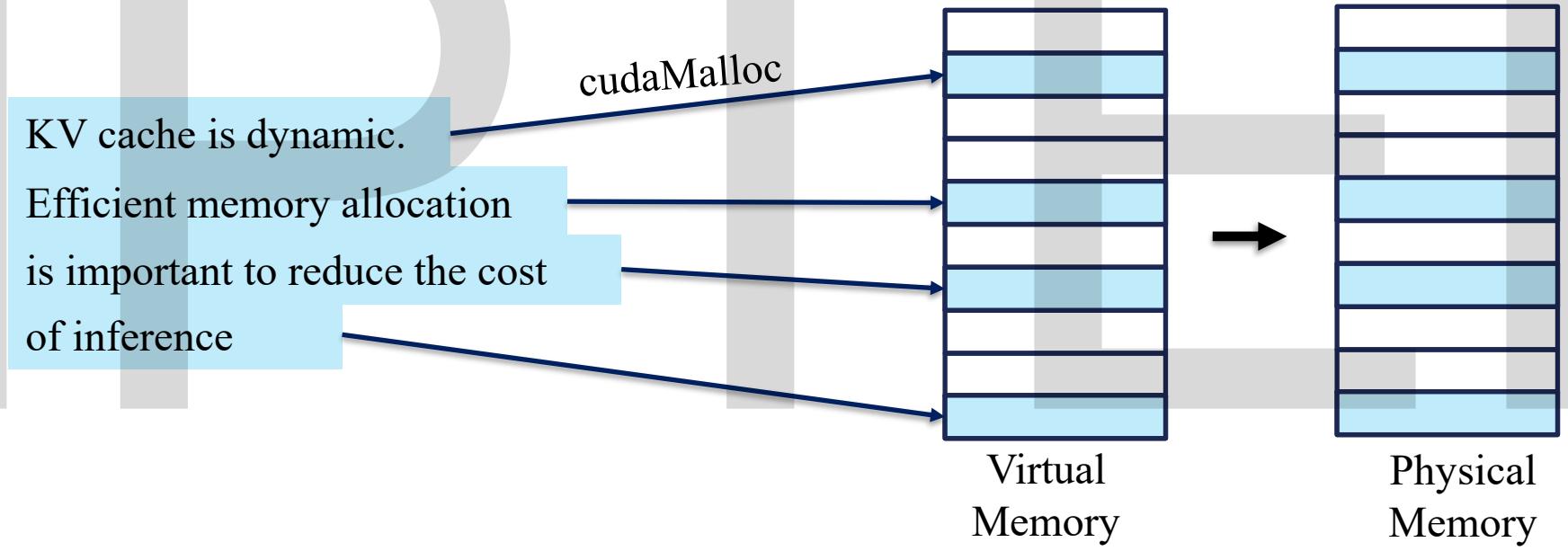
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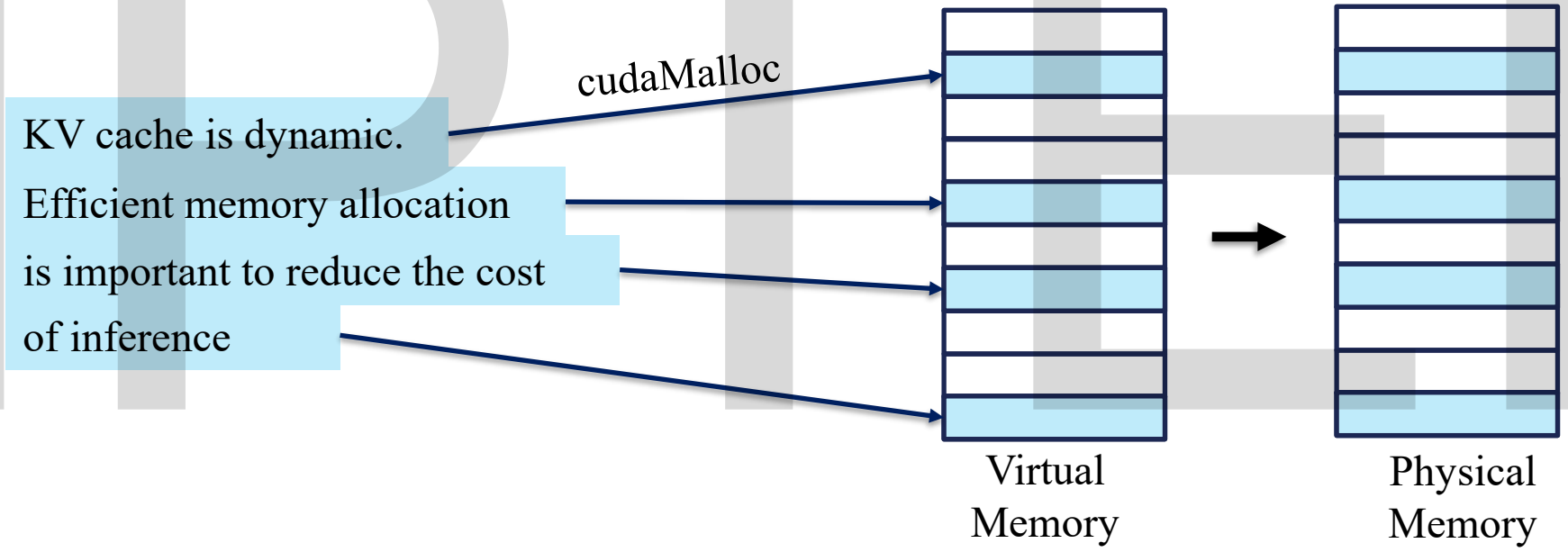
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Slides on vattention borrowed from Ashish Panwar

Summary

- **Key idea:** Make KV cache from a collection of small tensors (e.g., 16 tokens)



Pros:

- No fragmentation
- High batch size/throughput

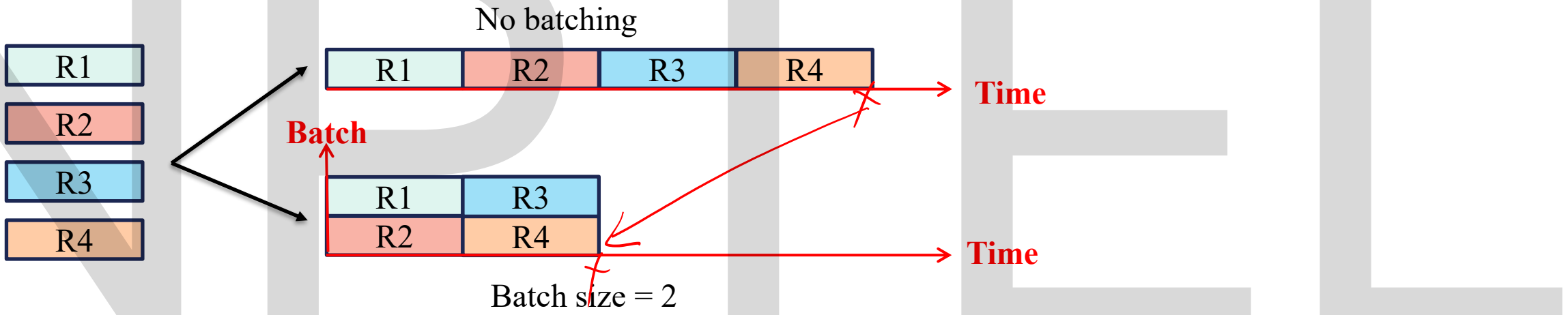
Slides on vattention borrowed from Ashish Panwar

Agenda

- Recap
- **Orca (LLM Serving)**

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Scheduling

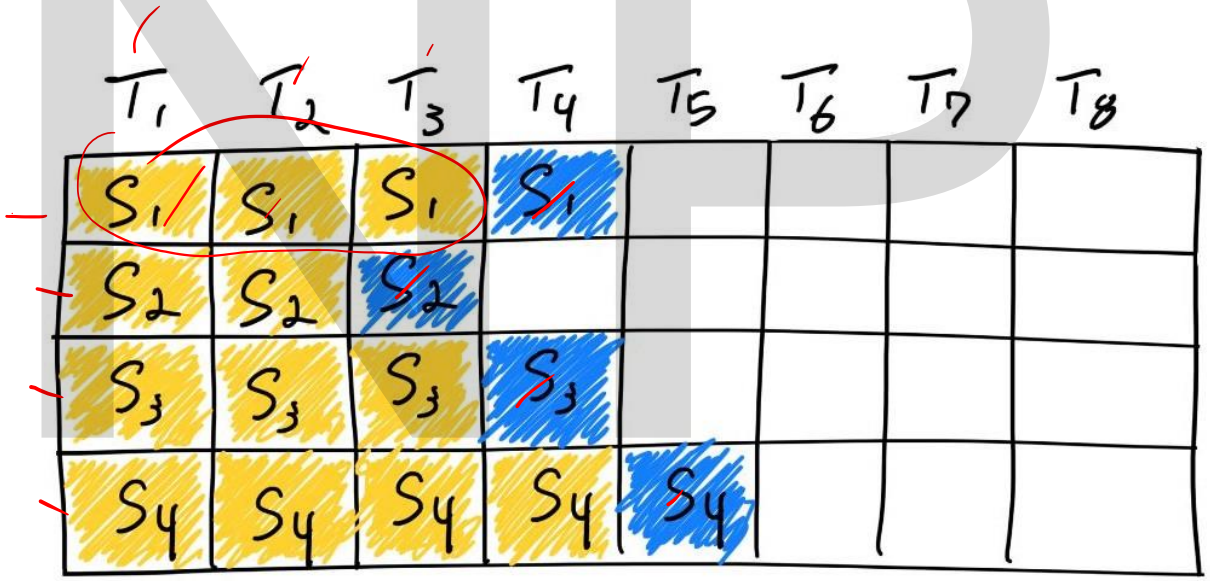


Why batch?

Exploit hardware parallelism, better serving efficiency

Static batching

- Batching multiple sequences on GPU, aka “static batching”



Problem: GPU utilization drops as sequences complete

Legend:

- Yellow: prompt token
- Blue: generated token
- Red: end-of-sequence token

source: AnyScale

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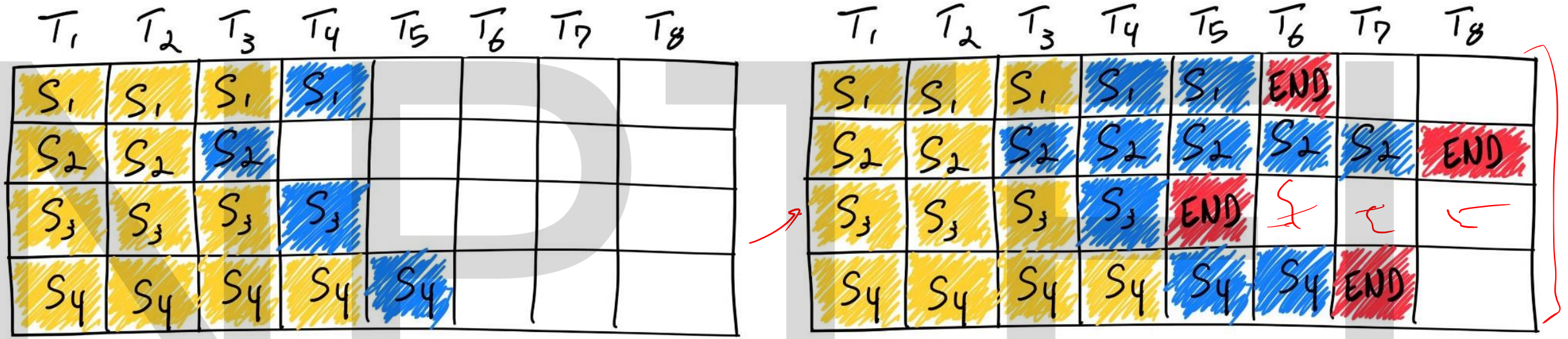
Continuous batching

Top: static batching
Bottom: continuous batching

Legend:

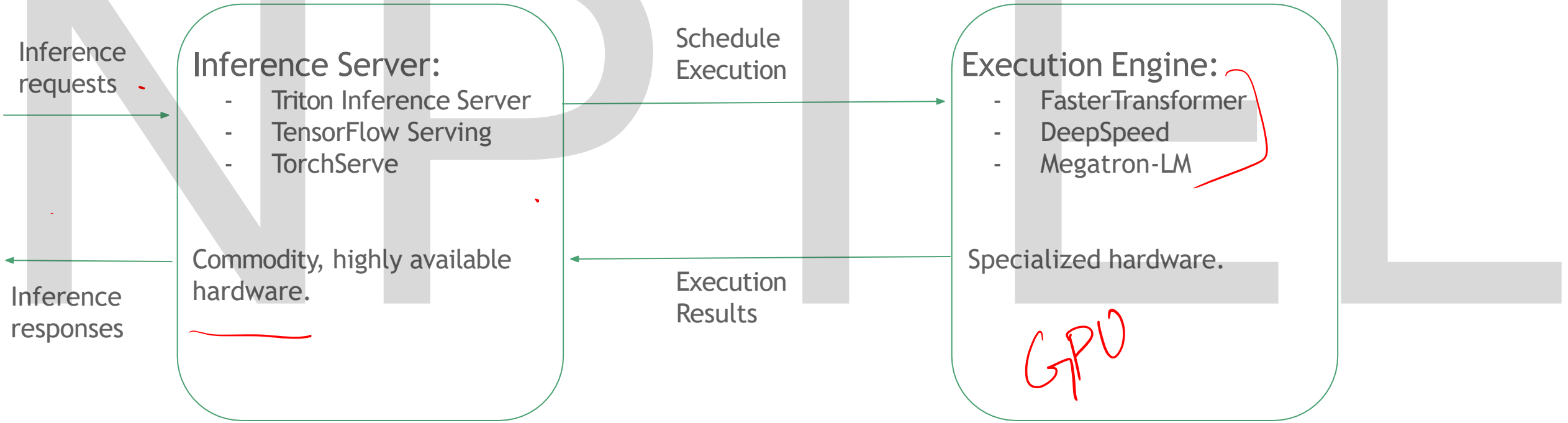
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Orca [OSDI 22]



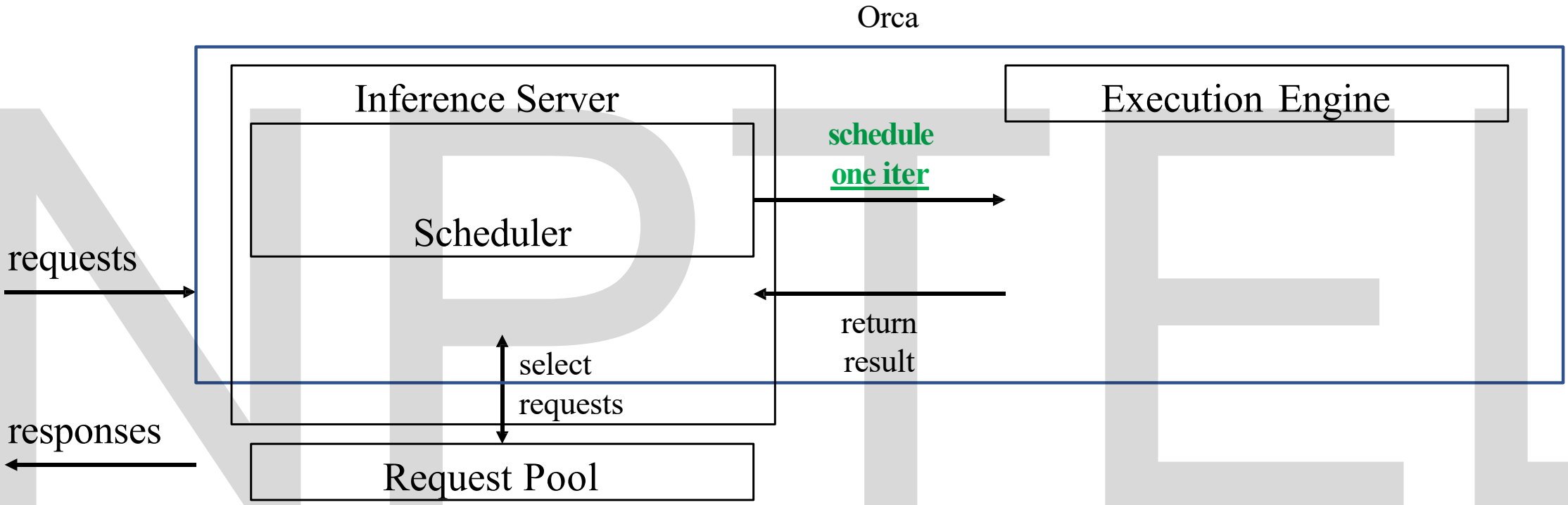
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Model Serving Architecture



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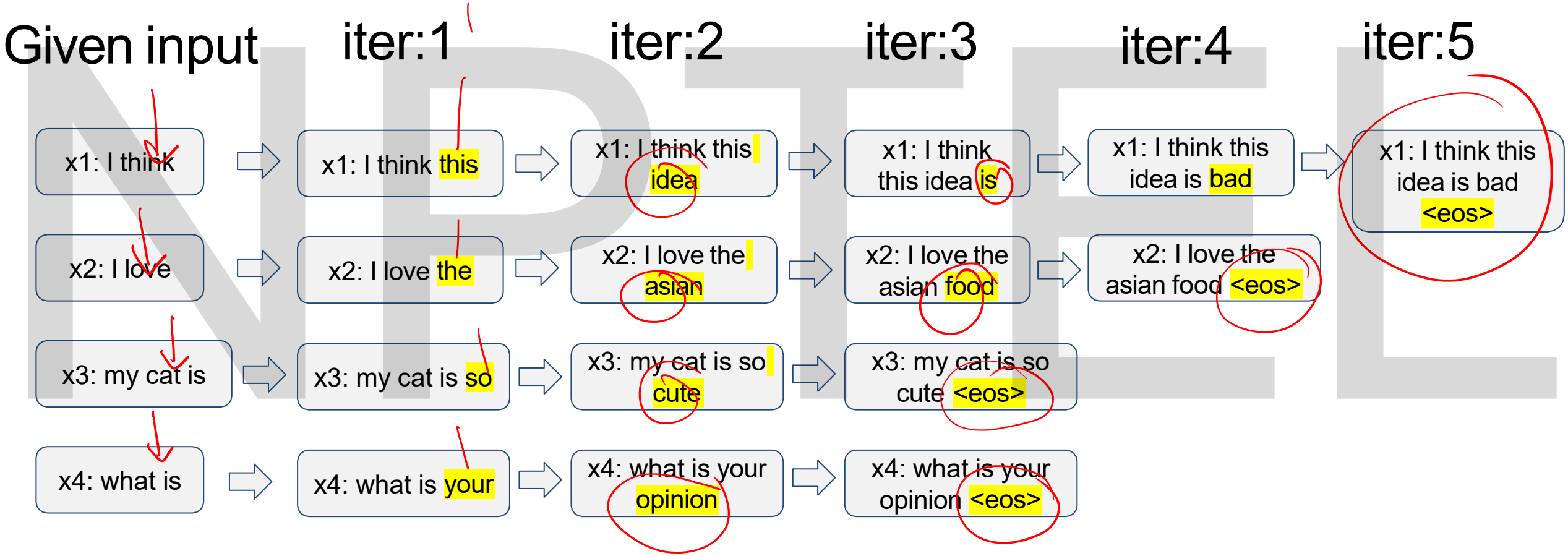
Solution : Iteration-Level Scheduling



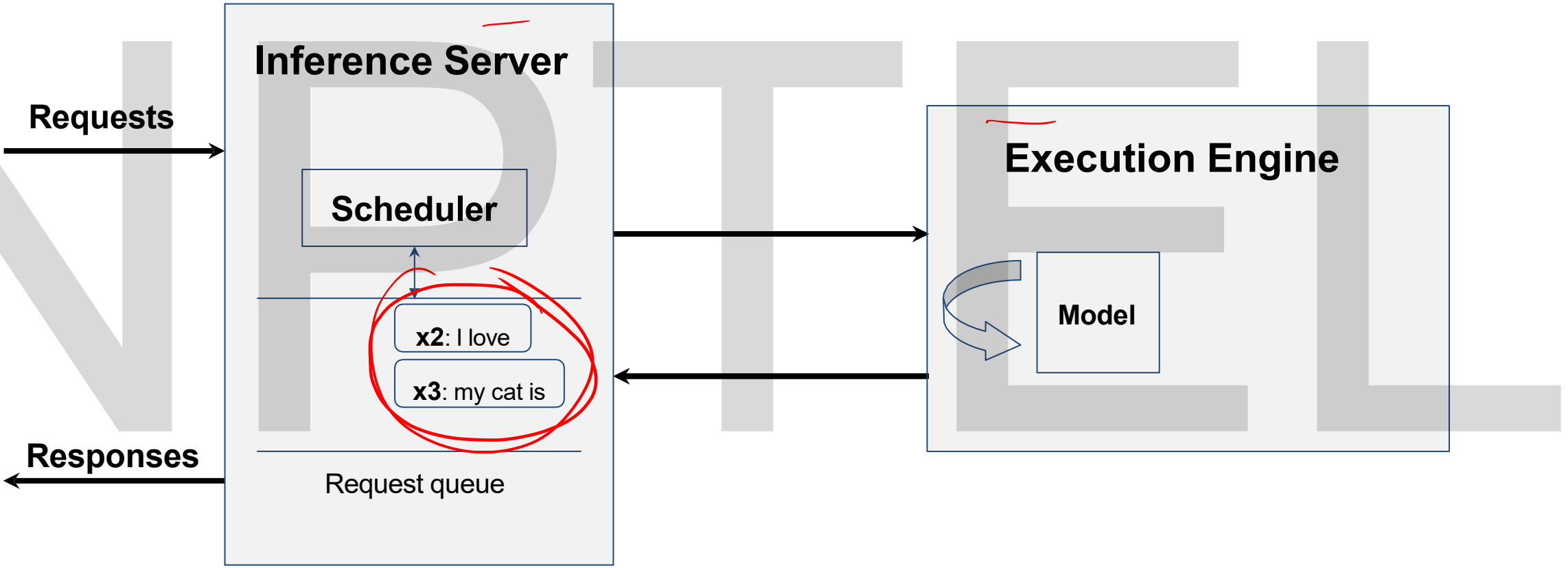
* maximum batch size = 3

source: Orca OSDI 2022 Talk

Example

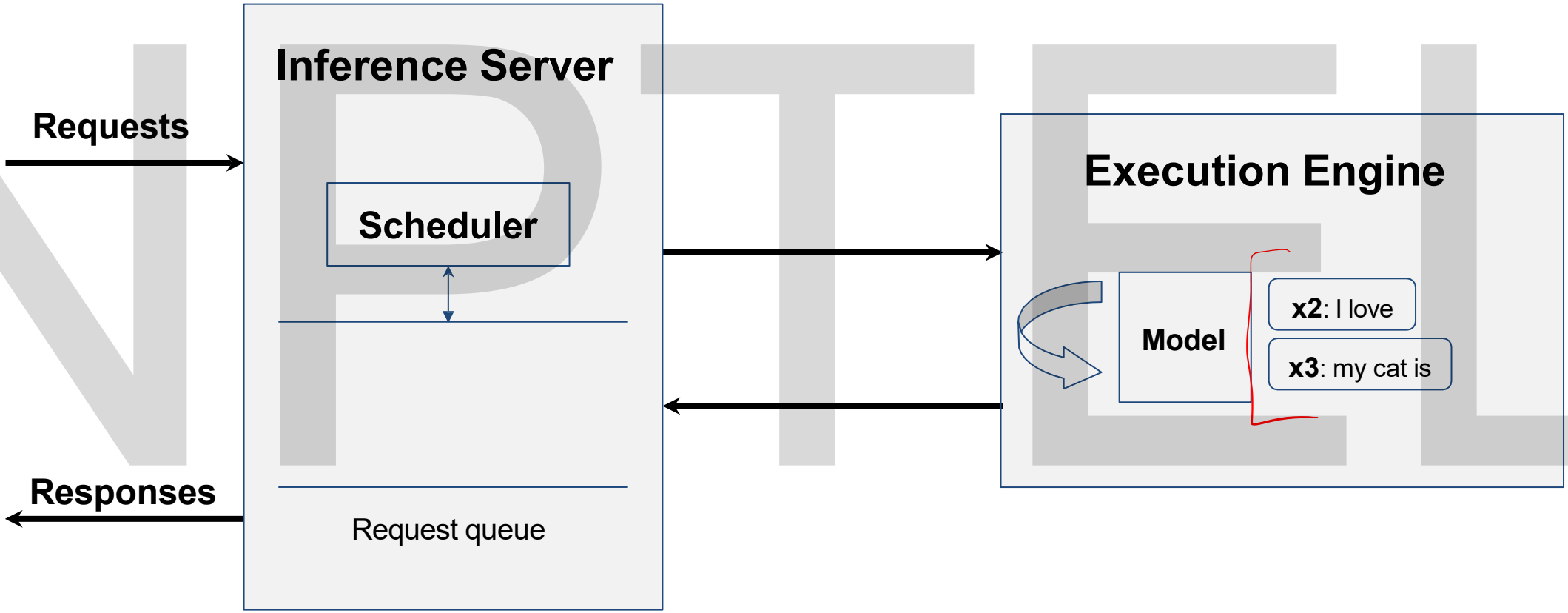

Given input
iter:1
iter:2
iter:3
iter:4
iter:5
x1: I think
x1: I think this
x1: I think this idea
x1: I think this idea is
x1: I think this idea is bad
x1: I think this idea is bad <eos>
x2: I love
x2: I love the
x2: I love the asian
x2: I love the asian food
x2: I love the asian food <eos>
x3: my cat is
x3: my cat is so
x3: my cat is so cute
x3: my cat is so cute <eos>
x4: what is
x4: what is your
x4: what is your opinion
x4: what is your opinion <eos>

Default : Request Scheduling



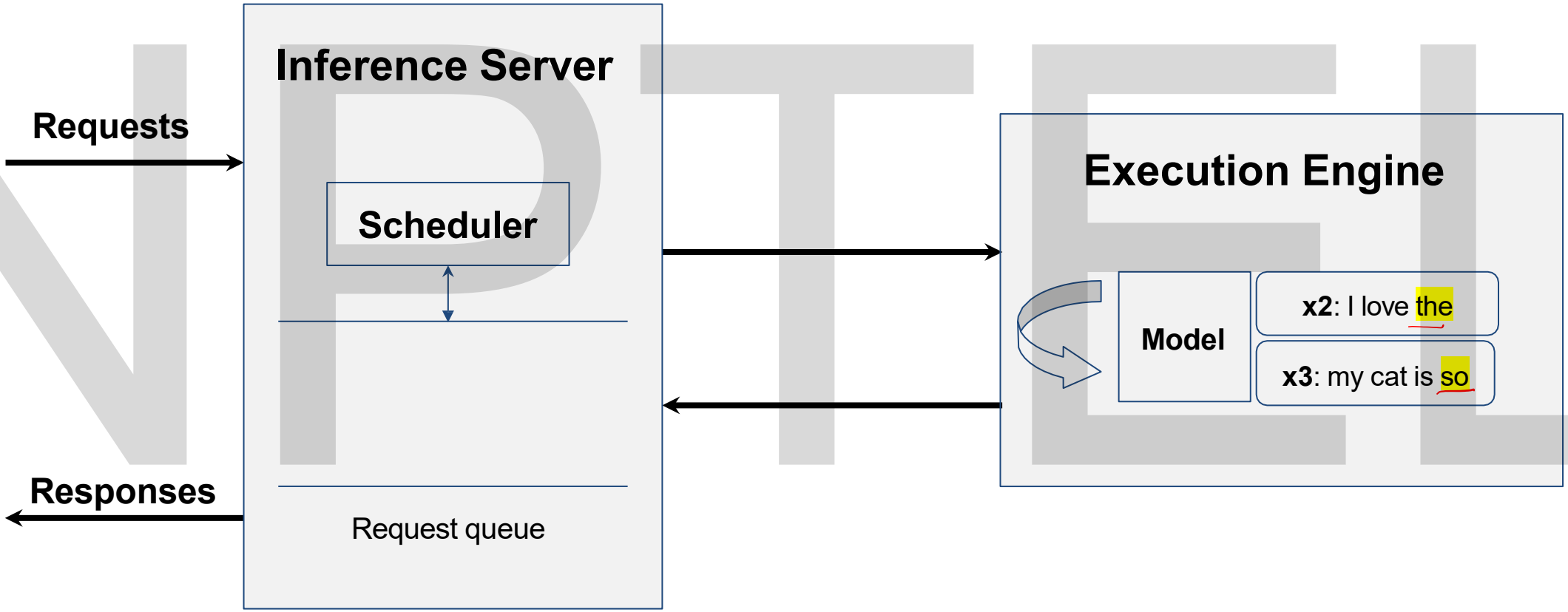
* Assume we set the max_batch_size = 3

Timestep 1



* Assume we set the max_batch_size = 3

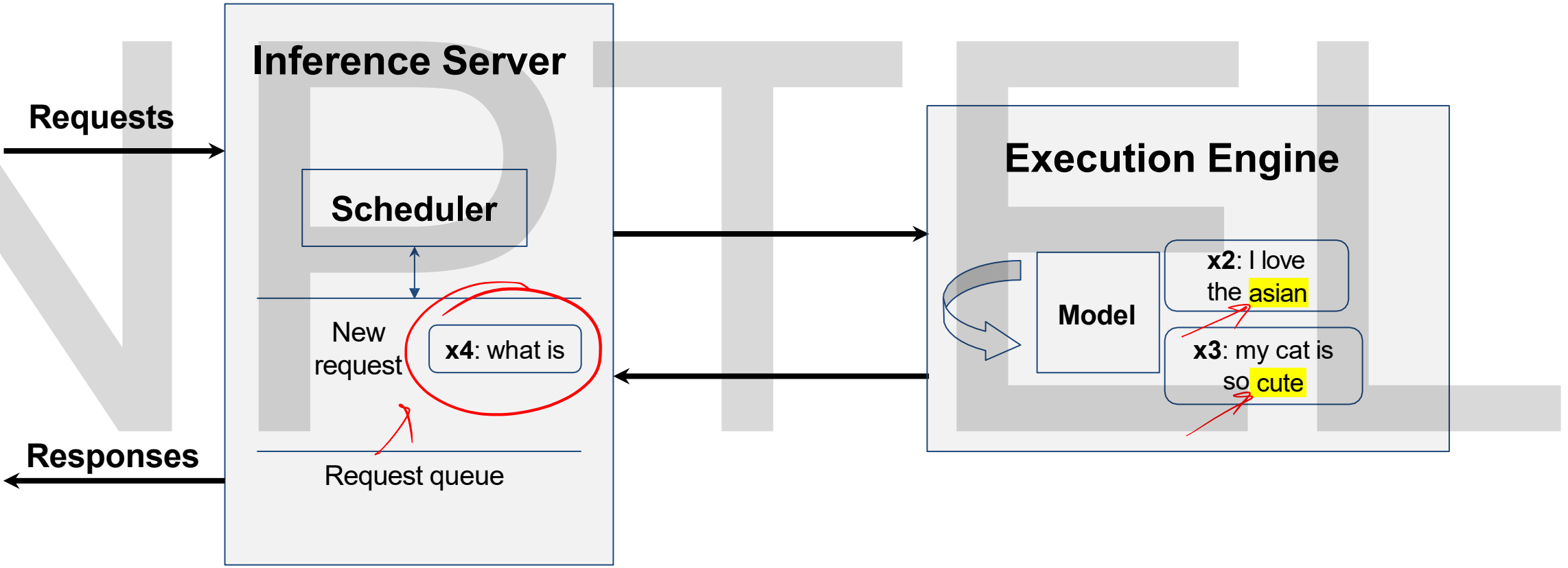
Timestep 2



*** Assume we set the max_batch_size = 3**

2
1

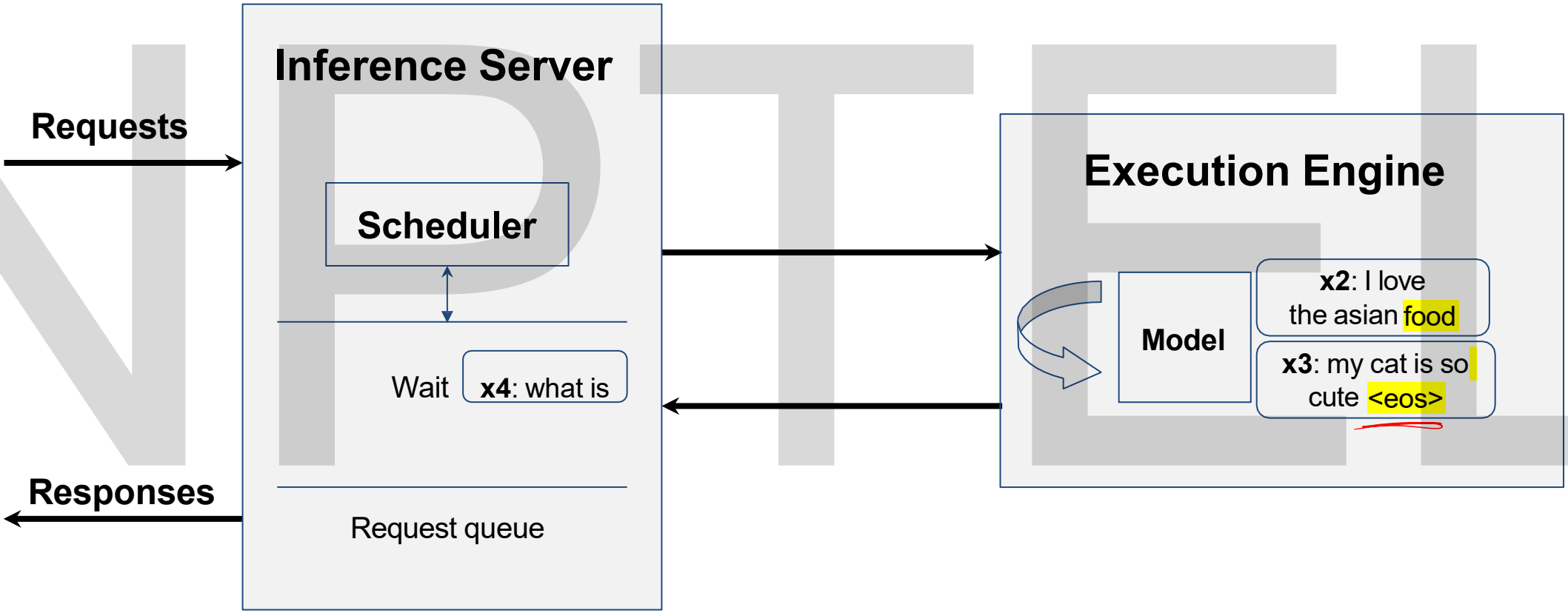
Timestep 3



*** Assume we set the max_batch_size = 3**

2
2

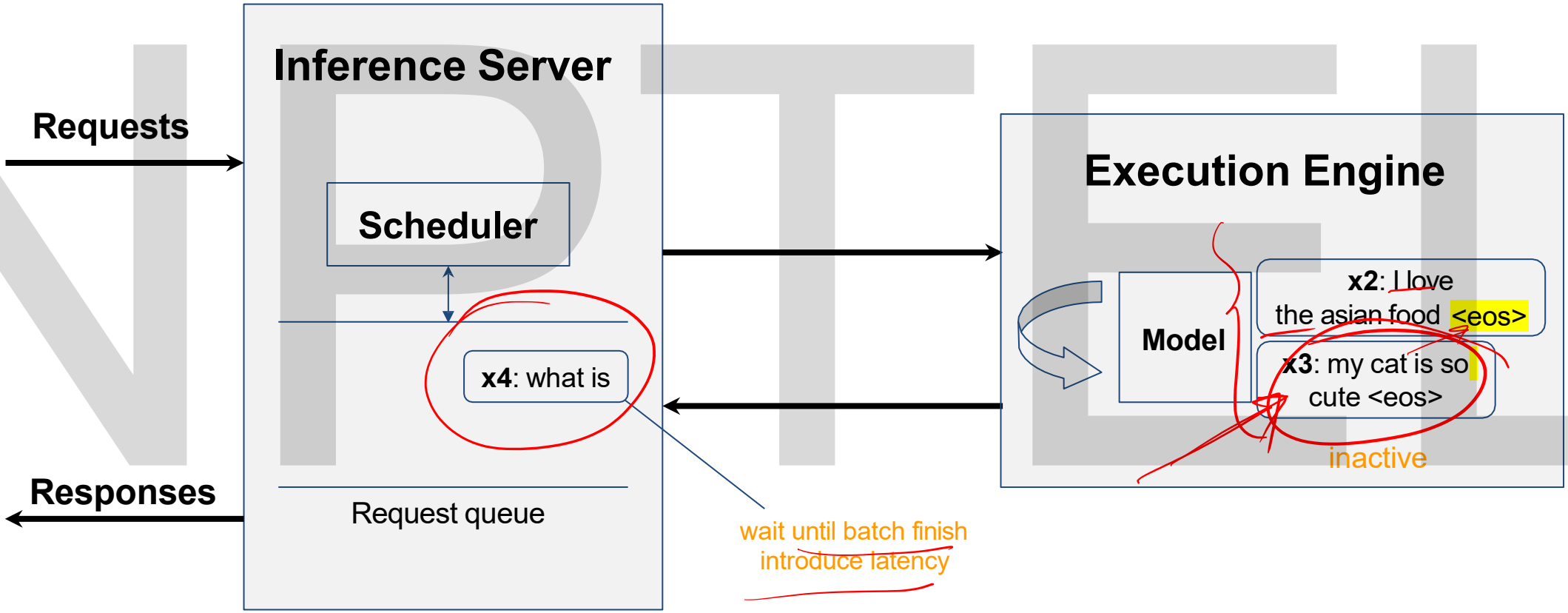
Timestep 4



*** Assume we set the max_batch_size = 3**

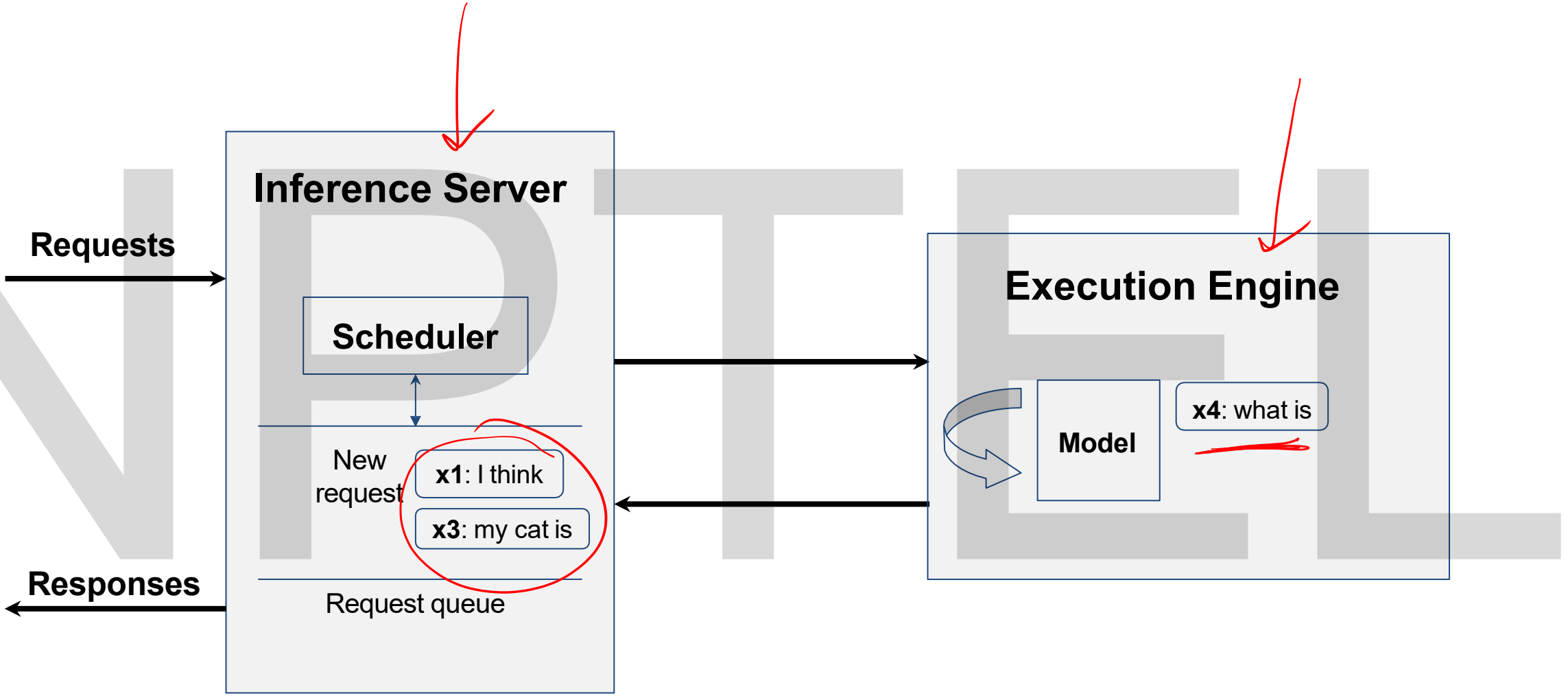
2
3

Timestep 5



*** Assume we set the max_batch_size = 3**

Timestep 6



* Assume we set the max_batch_size = 3

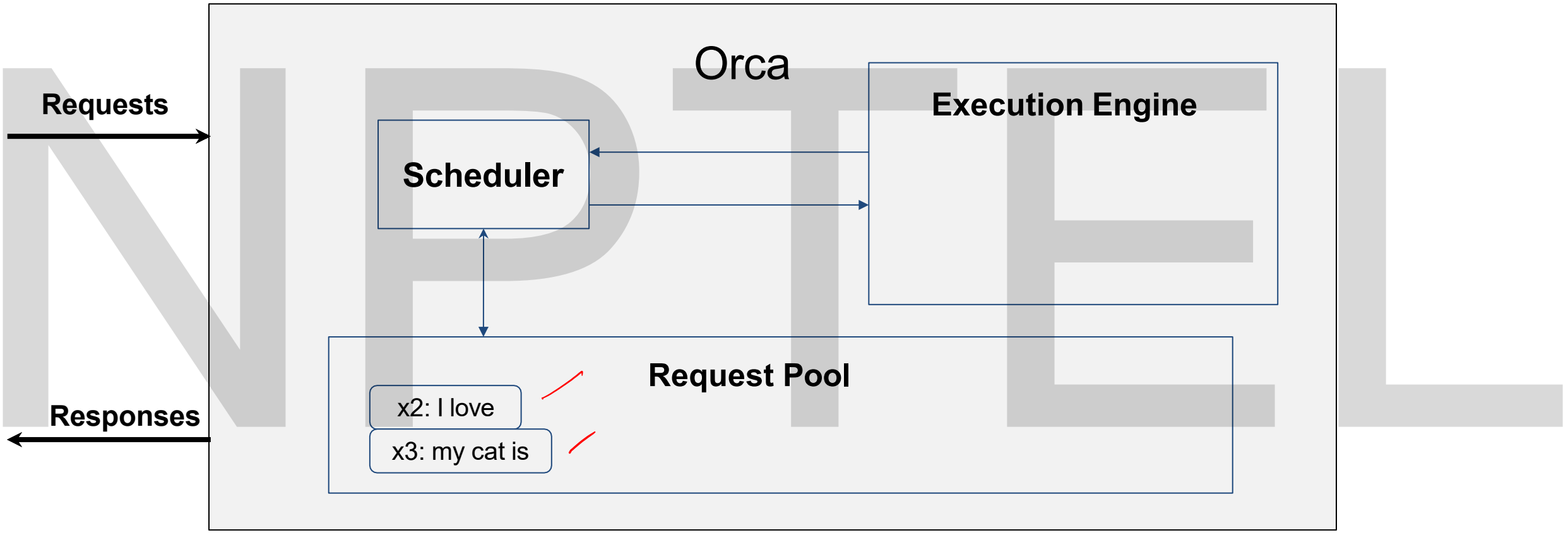
Drawback: High Latency

1. Low efficiency due to inactive requests

2. Newly arrived requests wait until all requests in the current batch have finished.

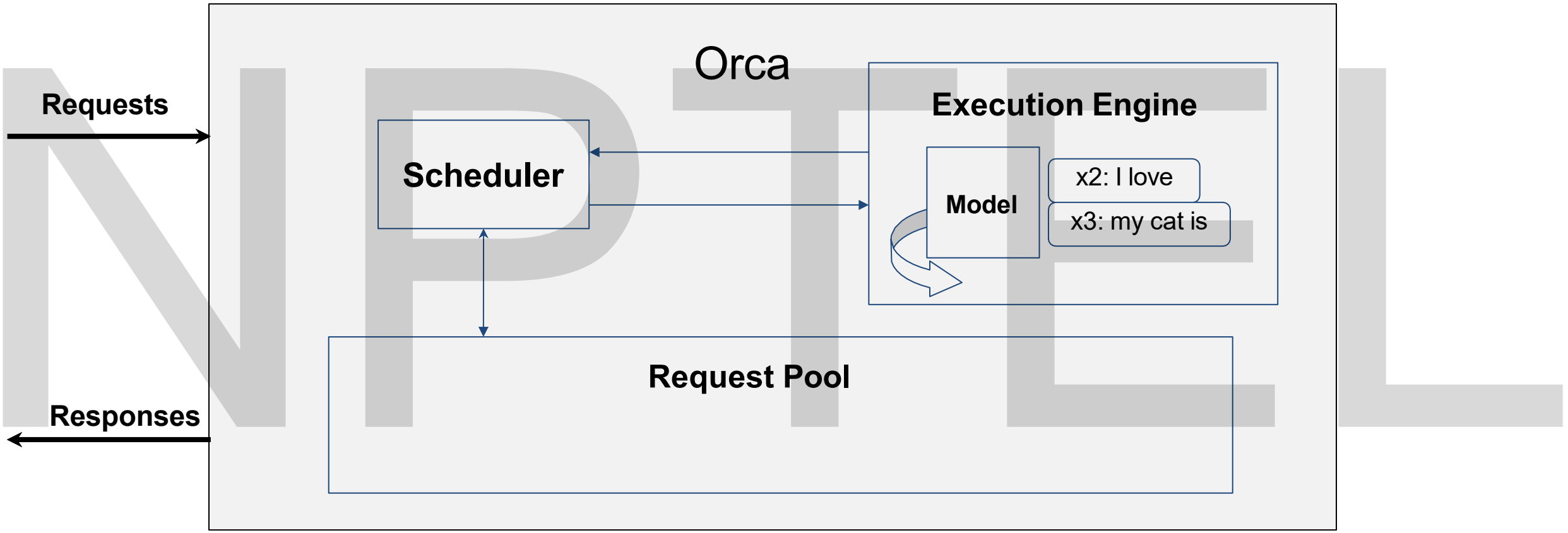
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Timestep 0



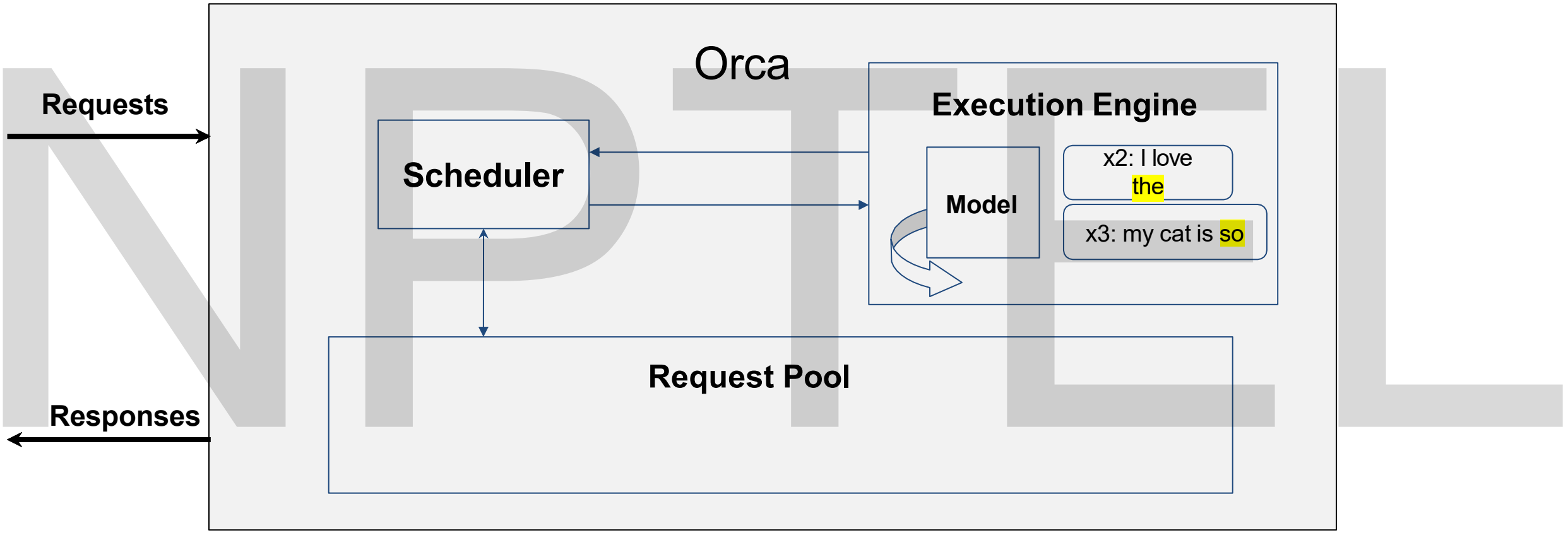
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Timestep 1



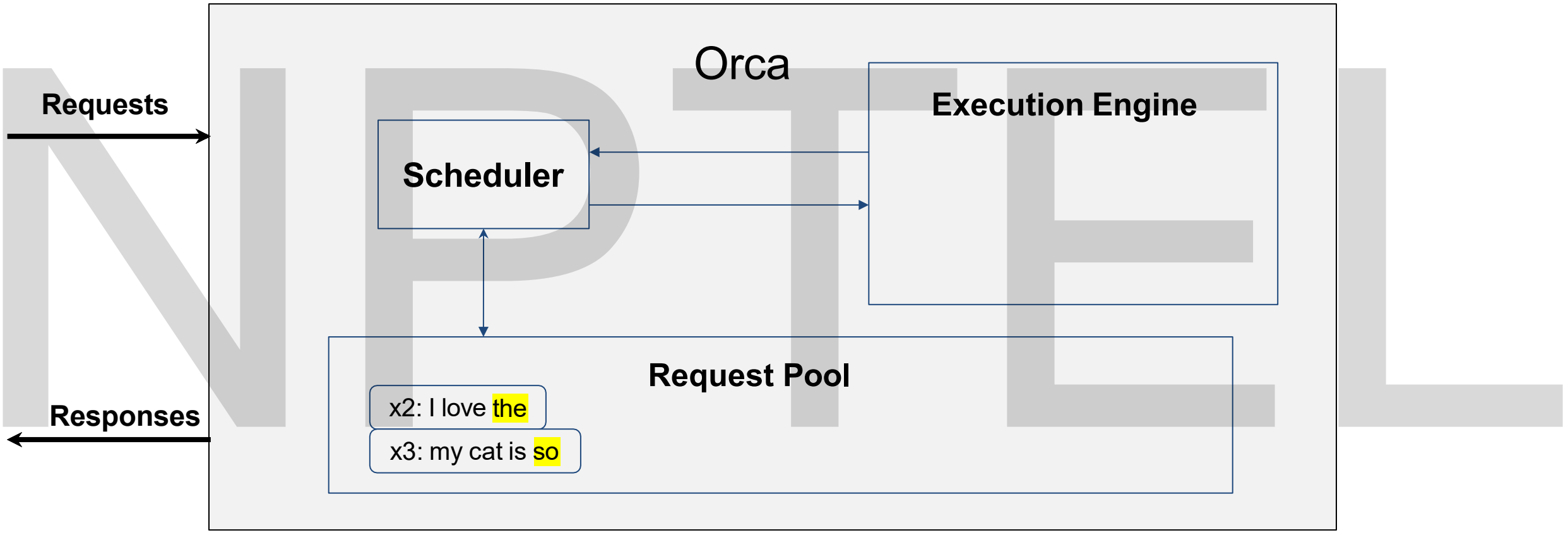
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Timestep 2



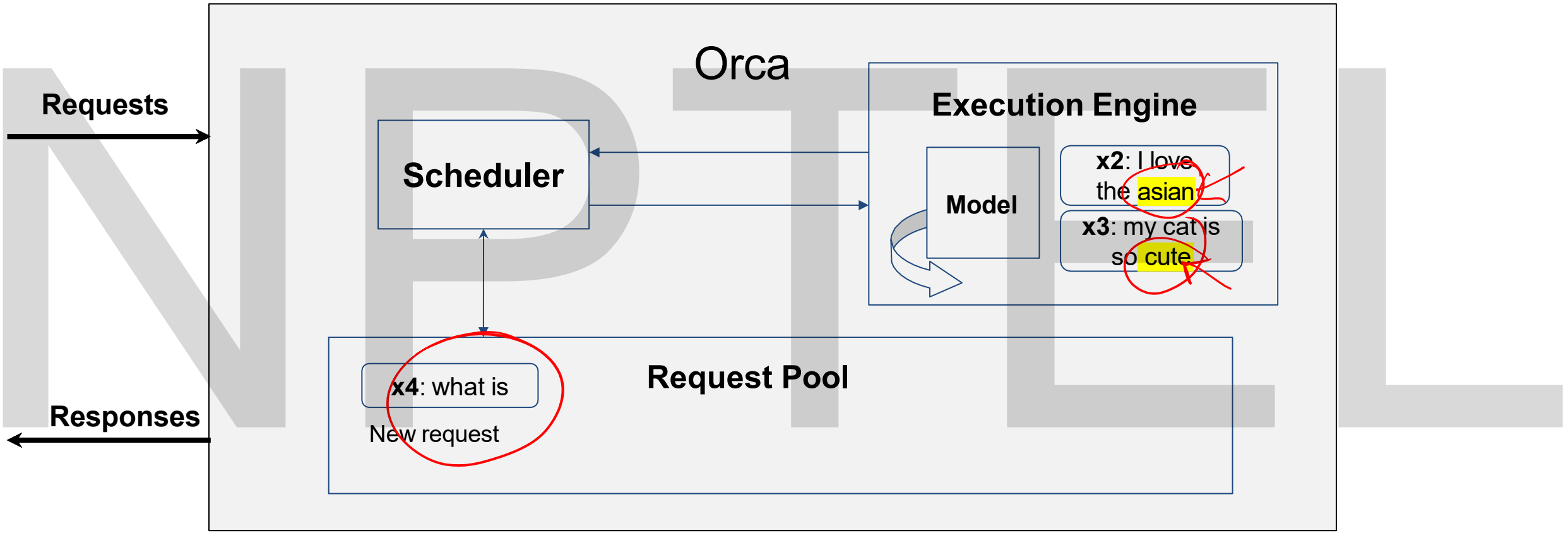
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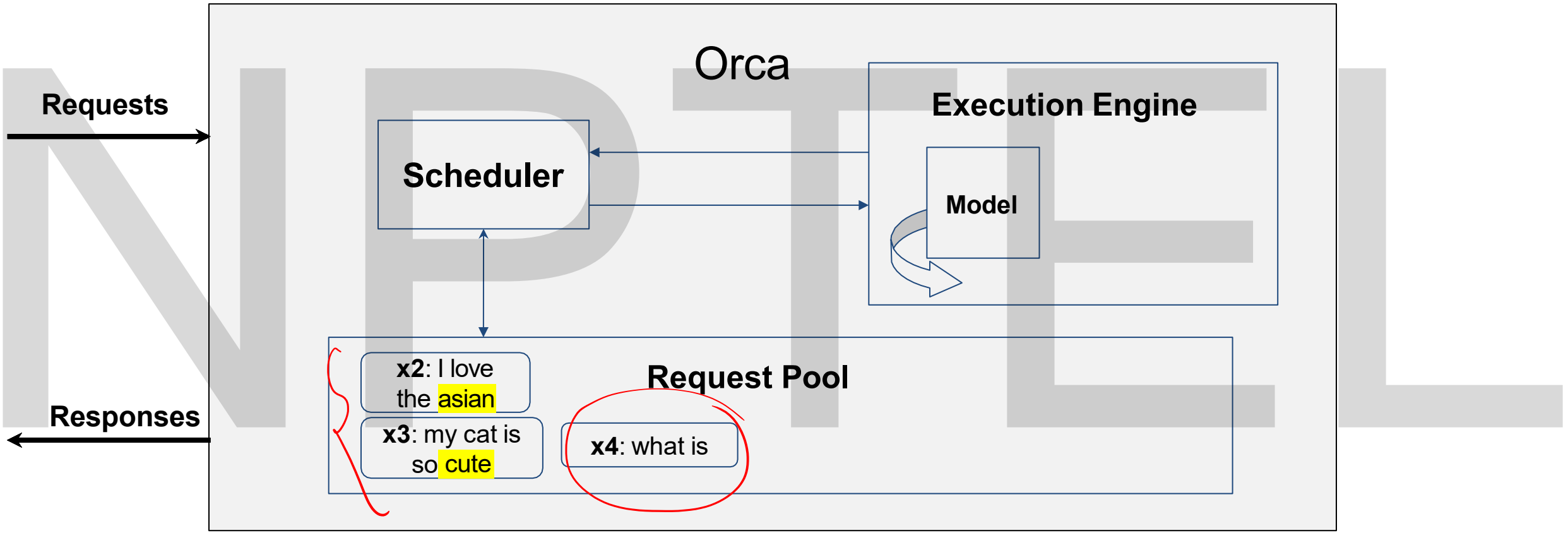
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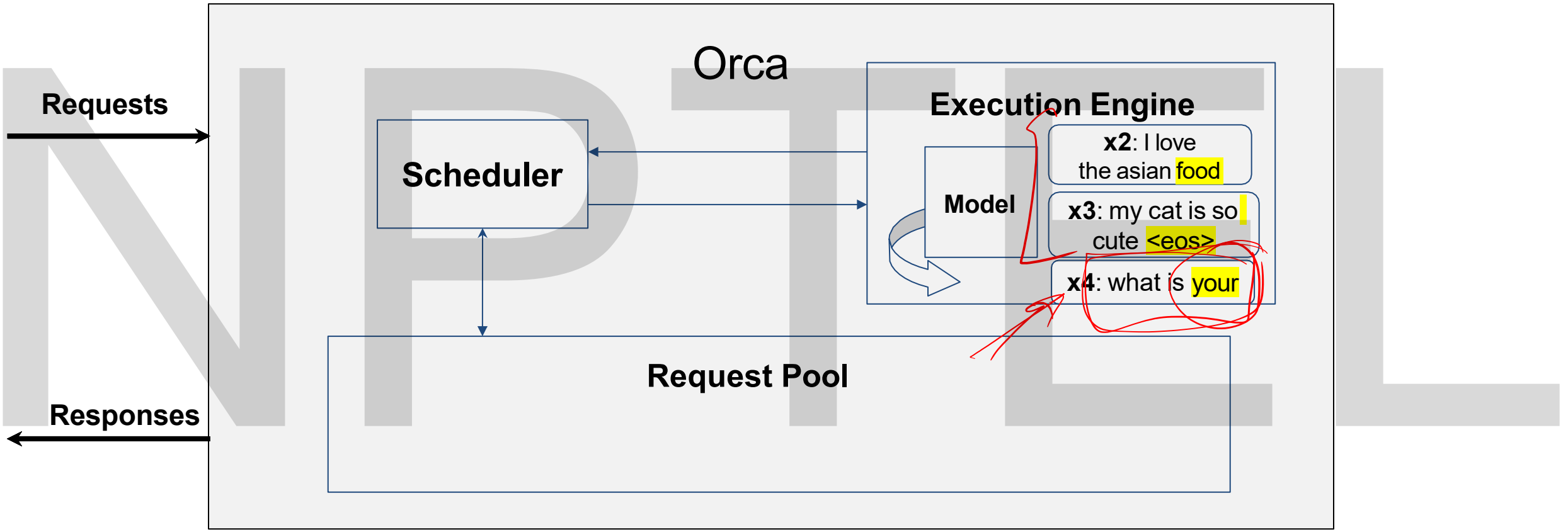
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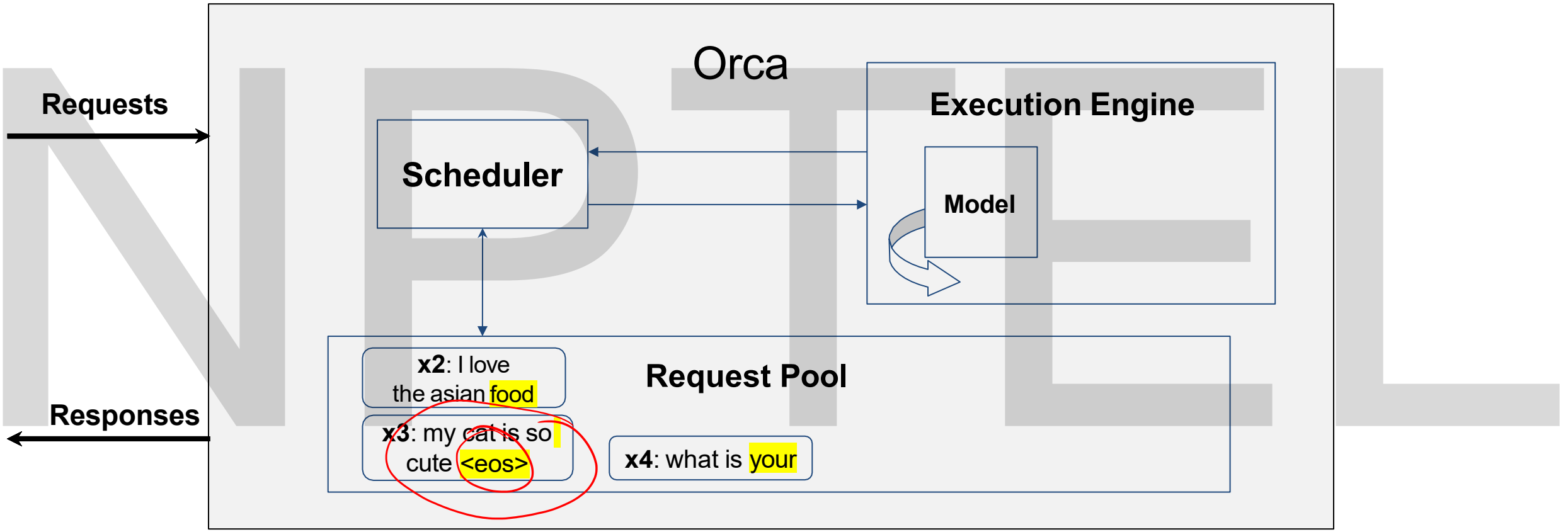
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Timestep 4



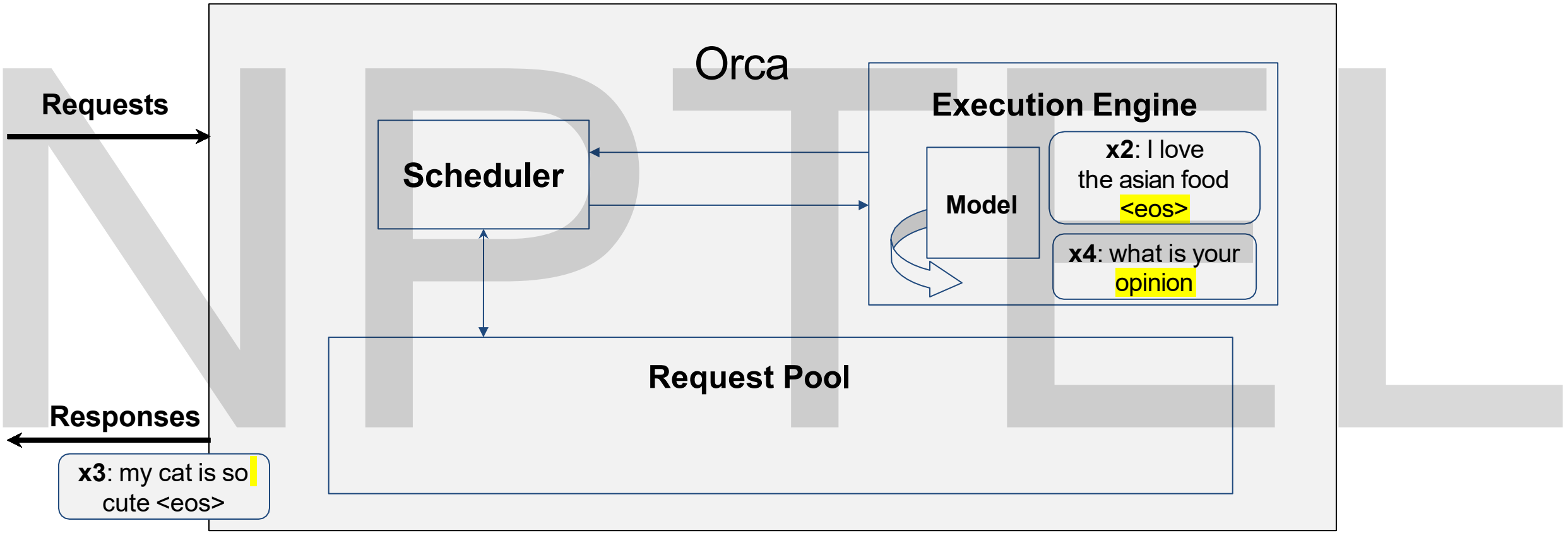
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Timestep 4



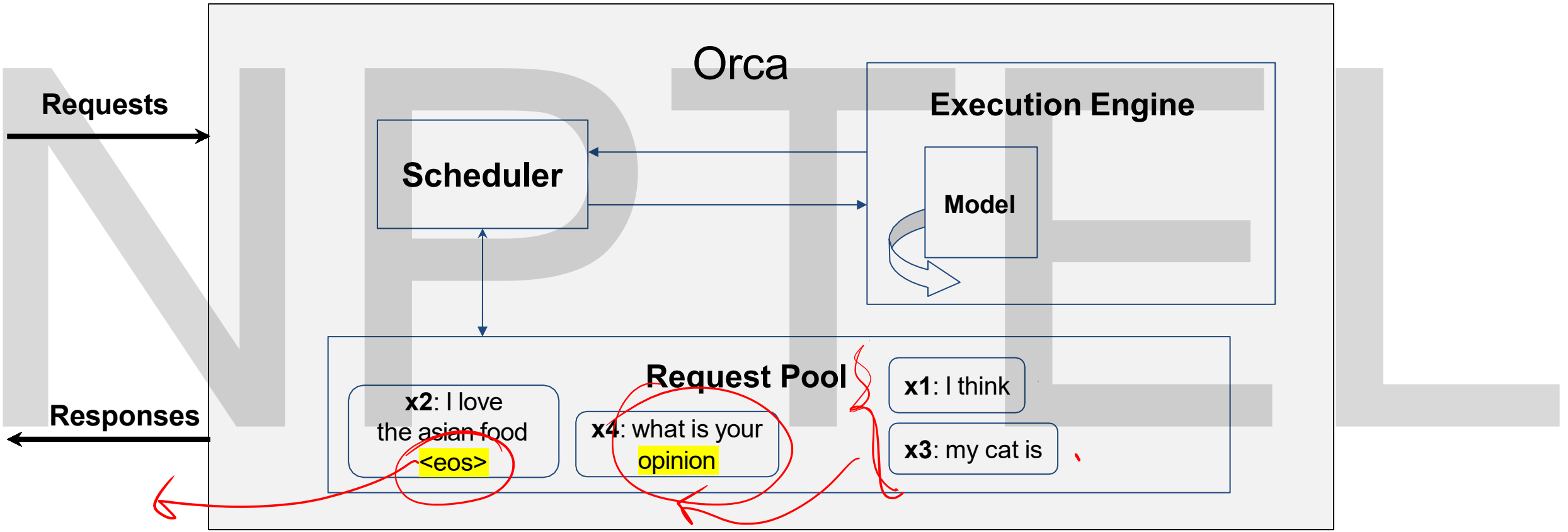
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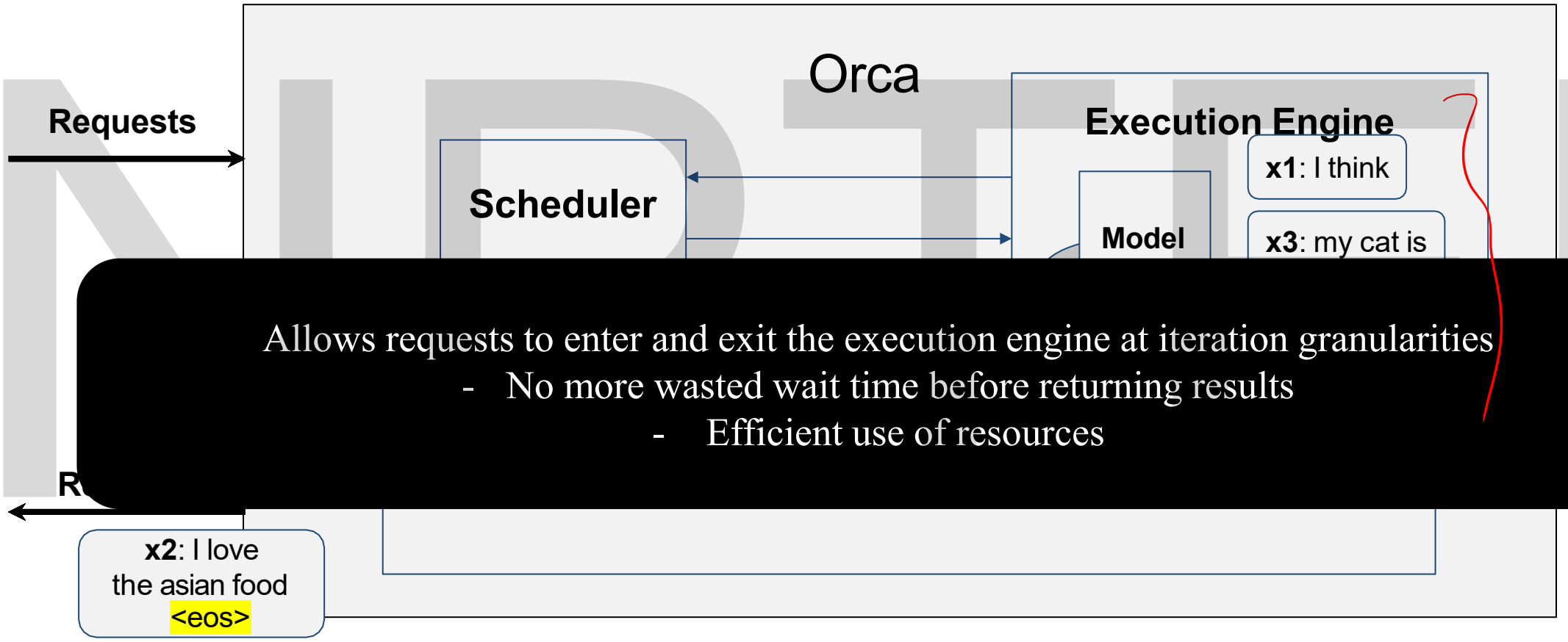
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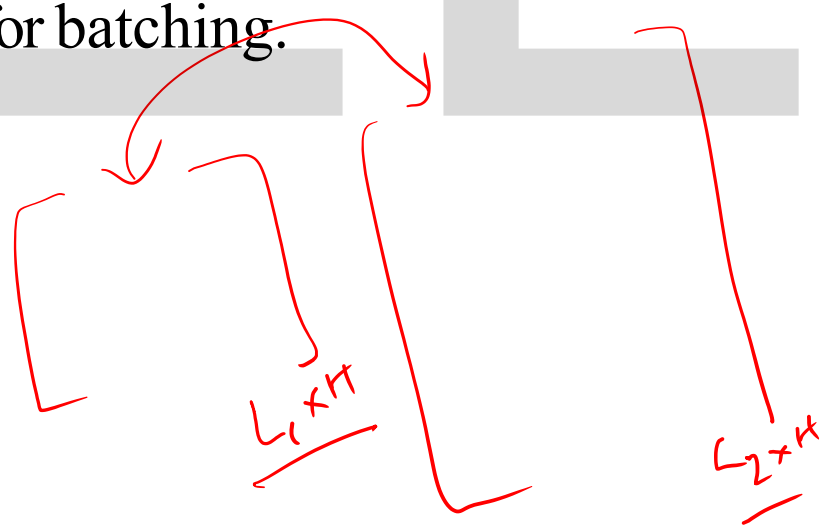
How to perform iteration-level batching?

- **To exploit parallelism with batching, all requests need to have the same number of tokens processed.**

- This is because the Attention mechanism requires input tensors to have the same shape (which is based on the number of processed tokens).

- Need to Coalesce B * [L, H]tensors into [B, L, H] tensor for batching.

  B -> batch size, L -> input length, H -> hidden size.



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How to perform iteration-level batching?

- **Batching is only applicable when**
  - **requests are in the same phase (prefill or decode)**
  - **requests have the same length**

X1: I think this is

X2: I love you

X3: A person

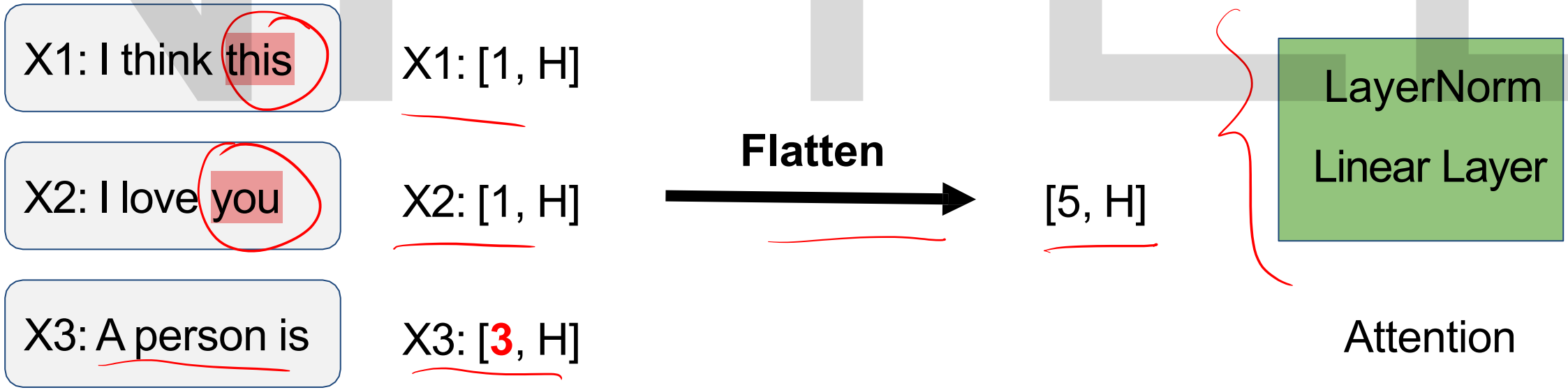
Three cases cannot batch normally:

1. both requests are in the prefill phase and each has different number of input tokens
2. both are in the decode phase and each is processing a token at different index from each other
3. each request is in the different phase: prefill or decode

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Orca Uses Selective Batching

Not all operations are incompatible with irregularly shaped tensors.
Matrix multiplication and layer normalization can be batched, because they do not distinguish different requests.



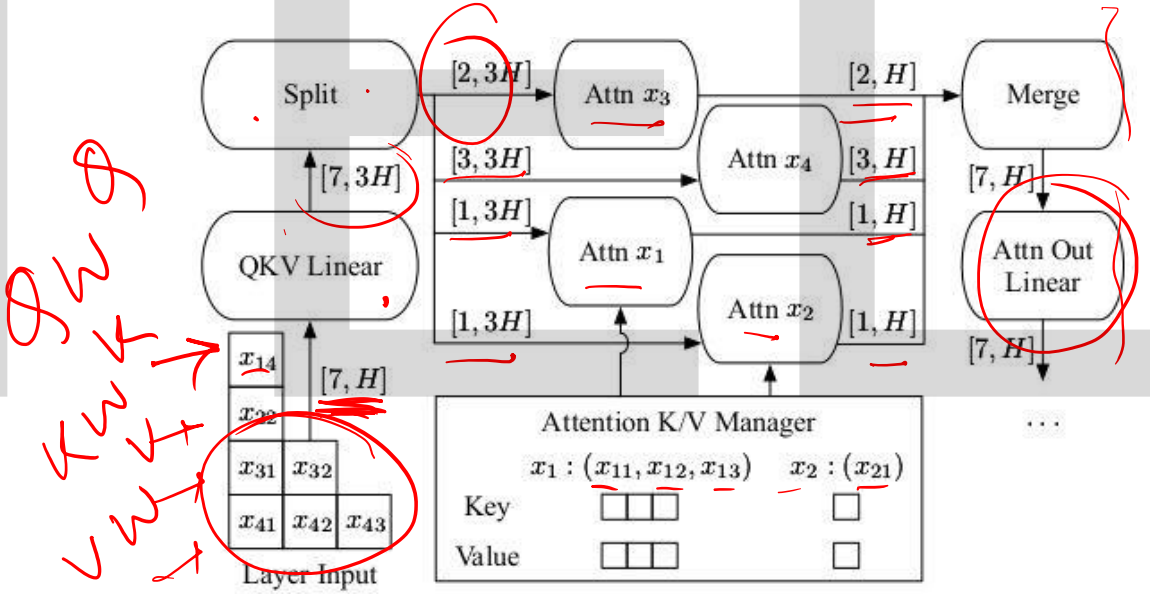
Solution: Selective Batching

Instead of batching **all** tensor operations, only batch the ones you can.

Flatten input tensors into 2-D tensor of shape [∑L, H], and feed into non-Attention operations (Linear, LayerNorm, Add, GeLU).

For Attention operations, split flattened tensor into individual requests and run Attention operation separately. Once completed, merge requests back.

Attention K/V manager provides Keys/Values separately for each request.





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)